Python Tuple Master Test

30 Multiple Choice Questions | Comprehensive Practice & Prep

Question 1

Which of the following syntaxes correctly creates a tuple containing a single element, integer 5?

A) t = (5)

B) t = (5,)

C) t = tuple(5)

D) t = [5]

Question 2

What is the primary characteristic that distinguishes a tuple from a list in Python?

A) Tuples use square brackets

B) Tuples are mutable

C) Tuples are immutable

D) Tuples cannot store heterogeneous data

Question 3

How many built-in methods do tuple objects natively have in Python?

A) 0

B) 2

C) 5

D) 11

Question 4

What is the output of the following code?

```
t = (10, 20, 30, 20, 40)
print(t.count(20))
```

A) 1

B) 2

C) 3

D) ValueError

Question 5

What happens when you run: t = (1, 2, 3); print(t.index(5))?

A) Returns -1

B) Returns None

C) Raises ValueError

D) Raises IndexError

Question 6

What is the output of the code below?

```
t = (10, 20, 30, 40, 50)
print(t[1:4])
```

A) (20, 30, 40)

B) (10, 20, 30)

C) (20, 30, 40, 50)

D) [20, 30, 40]

Question 7

Which of the following options will result in a TypeError?

A) t = (1, 2, 3) + (4, 5)

B) t = (1, 2) * 3

C) t = (1, 2, 3); t[0] = 10

D) print(len((1, 2)))

Question 8

What is the output of the following snippet?

```
t = (1, [2, 3], 4)
t[1][0] = 99
print(t)
```

A) TypeError

B) (1, [99, 3], 4)

C) (1, [2, 3], 4)

D) AttributeError

Question 9

What is the output of the code?

```
a, b, *c = (1, 2, 3, 4, 5)
print(c)
```

A) (3, 4, 5)

B) [3, 4, 5]

C) 3, 4, 5

D) (1, 2)

Question 10

What does the built-in function tuple('python') return?

A) ('python',)

B) ('p', 'y', 't', 'h', 'o', 'n')

C) ['p', 'y', 't', 'h', 'o', 'n']

D) Error

Question 11

What is the result of evaluating: (1, 2) < (1, 3)?

A) True

B) False

C) TypeError

D) None

Question 12

What is the return type of the sorted() function when applied to a tuple, e.g., sorted((3, 1, 2))?

A) Tuple

B) List

C) Set

D) Generator

Question 13

What is the output of the following code?

```
t = (1, 2, 3)
print(type(t))
```

A)

B)

C)

D)

Question 14

Which of the following is true regarding tuple keys in a Python dictionary?

A) Tuples can never be used as dictionary keys

B) Any tuple can be used as a dictionary key

C) Tuples can be keys only if all their elements are hashable

D) Lists are preferred over tuples as dictionary keys

Question 15
What will be the output of: print(bool(()))?

A) True

B) False

C) None

D) TypeError

Question 16
What is the output of the following code?

```
marks = (56, [78, 90], 89)
print(len(marks))
```

A) 3

B) 4

C) 5

D) TypeError

Question 17
How can you find the index of the second occurrence of element 10 in t = (10, 20, 10, 30, 10)?

A) t.index(10, 1)

B) t.index(10, 2)

C) t.index(10, 0, 2)

D) t.count(10)[1]

Question 18
What happens when you execute: del t[0] on t = (1, 2, 3)?

A) Removes 1 from tuple

B) Raises TypeError

C) Returns (2, 3)

D) Deletes entire tuple

Question 19
Which expression checks if 25 is present in the tuple age = (18, 21, 25, 30)?

A) 25 in age

B) age.contains(25)

C) age.has(25)

D) 25.exists(age)

Question 20
What is the output of: print((1, 2) + (3, 4))?

A) (4, 6)

B) (1, 2, 3, 4)

C) [1, 2, 3, 4]

D) Error

Question 21
What is the result of memory footprint comparison between a list and tuple of identical elements?

A) List uses less memory than tuple

B) Tuple uses less memory than list

C) Both use equal memory

D) Depends on Python version

Question 22
What is the output of: t = (10,); print(type(t))?

A)

B)

C)

D)

Question 23
What is the output of: print(min(('apple', 'Banana', 'cherry')))?

A) 'apple'
B) 'Banana'
C) 'cherry'
D) TypeError

Question 24
Which operator is used for tuple unpacking?

A) %
B) Assignment operator (=)
C) &
D) @

Question 25
What is the output of: t = (1, 2, 3, 4); print(t[-1])?

A) 1
B) 3
C) 4
D) IndexError

Question 26
What does zip([1, 2], ('a', 'b')) produce when converted to a tuple?

A) ((1, 2), ('a', 'b'))
B) ((1, 'a'), (2, 'b'))
C) [1, 'a', 2, 'b']
D) Error

Question 27
What is the output of: print((0,) * 4)?

A) 0
B) (0, 0, 0, 0)
C) (0, 4)
D) [0, 0, 0, 0]

Question 28
What is the output of: t = (10, 20, 30); print(30 in t)?

A) True
B) False
C) 2
D) None

Question 29
Which statement correctly creates an empty tuple?

A) t = ()
B) t = tuple()
C) Both A and B
D) Neither A nor B

Question 30
What is the output of: t = (1, 2, 3); print(sum(t, 10))?

A) 6
B) 16
C) TypeError
D) 10

Answer Key & Detailed Explanations

Q#	Answer	Explanation

Q1	**B**	A trailing comma is required for single-element tuples, e.g., (5,). Without a comma, (5) is evaluated as a simple integer inside parentheses.
Q2	**C**	Tuples are immutable, meaning their elements cannot be changed, added, or removed once created.
Q3	**B**	Tuple objects only have 2 built-in methods: count() and index().
Q4	**B**	The count() method returns the number of times the value appears in the tuple (20 appears twice).
Q5	**C**	If the specified item is not found in the tuple, index() raises a ValueError.
Q6	**A**	Slicing from index 1 up to (but not including) index 4 yields elements at indices 1, 2, and 3: (20, 30, 40).
Q7	**C**	Item assignment is prohibited on tuples because they are immutable.
Q8	**B**	Although the tuple itself is immutable, the list element inside it is mutable and can be modified in place.
Q9	**B**	Extended tuple unpacking collects remaining items into a list assigned to *c.
Q10	**B**	Passing a string to tuple() converts each character into a tuple element.
Q11	**A**	Tuple comparison compares elements position by position. Since 1 == 1, it checks 2 < 3, which is True.
Q12	**B**	The built-in sorted() function always returns a new list.
Q13	**B**	The type of a tuple object is .
Q14	**C**	A tuple can serve as a dict key only if it contains immutable/hashable elements (e.g. no lists inside).
Q15	**B**	An empty tuple () evaluates to False in boolean context.
Q16	**A**	The top-level tuple has 3 elements: integer 56, list [78, 90], and integer 89.
Q17	**A**	t.index(10, 1) starts searching from index 1, finding the next 10 located at index 2.
Q18	**B**	Deleting items using 'del t[0]' is not supported on tuples and raises TypeError.
Q19	**A**	The 'in' operator checks for membership in sequence structures like tuples.
Q20	**B**	concatenation (+) combines two tuples into a new single tuple.
Q21	**B**	Tuples are immutable and smaller/more memory-efficient than lists in Python.
Q22	**B**	The comma makes (10,) a tuple object.
Q23	**B**	String comparison is ASCII-based. Uppercase 'B' (66) comes before lowercase 'a' (97).
Q24	**B**	Tuple unpacking assigns tuple elements to variables across an assignment (=) statement, e.g., x, y = (1, 2).
Q25	**C**	Negative indexing -1 accesses the last element of the tuple.
Q26	**B**	zip pairs elements position-wise: (1, 'a') and (2, 'b').
Q27	**B**	The multiplication operator (*) repeats the tuple 4 times.
Q28	**A**	30 is present in the tuple, so '30 in t' returns True.
Q29	**C**	Both () literal notation and tuple() constructor create empty tuples.
Q30	**B**	sum(iterable, start) adds elements of iterable to the 'start' value (10 + 1 + 2 + 3 = 16).